Cart lab: change mass

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UNIT 1 / DAY 11 / CPA + SEI

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a controlled comparison to describe the effect of system mass on acceleration.

SUCCESS CRITERIA - I can...

1. Identify total moving mass and keep driving force fixed.
2. Compare acceleration values using a fair test.
3. Separate supplied or simulated values from values you measured.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mass” and “system” to explain your answer. Sentence starters are optional.

mass: a measure of how strongly an object resists acceleration

system: the object or objects chosen for analysis

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A cart accelerates uniformly from rest to **12 m/s** *east* in **4 s**. **Find acceleration**, average velocity and displacement.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

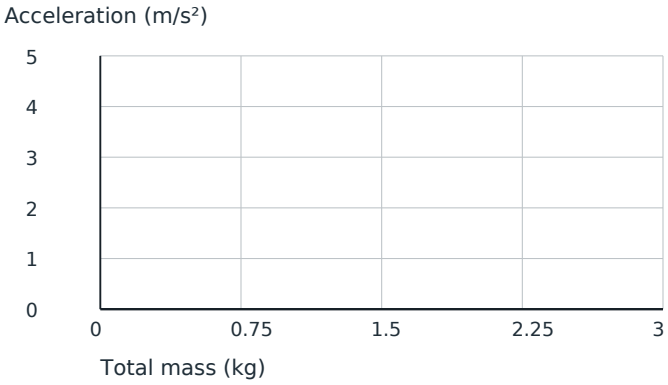
02 Worked example: explain the prediction

Reference: $a = F_{net}/M = m_h\ g/M$.
Ideal cart + hanger: M is all moving mass; m_h is hanging mass. Hold m_h fixed; neglect resistance and pulley inertia.

Worked reasoning: Supplied cart trials use the same net force: masses **1** and **2 kg** accelerate at **4** and **2 m/s²**. In both cases ma=**4 N**; doubling mass halves acceleration.

Your turn: Plot acceleration against mass for the two trials. **Label what** was held constant.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

03 Read the sample pattern

The table gives ideal cart trials with the same net force of **4 N**. **Compare the 1 kg** and **2 kg** trials. **What happens** to acceleration when mass doubles? **Use both acceleration** values to **support your answer**.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .

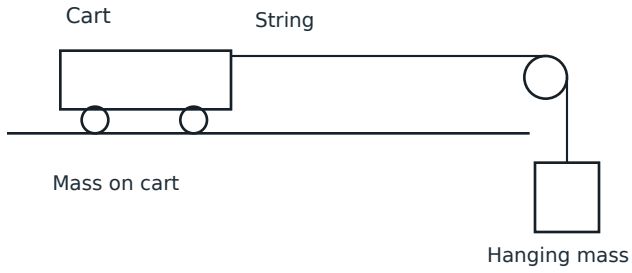
M (kg)	a (m/s²)
1	4
2	2
4	1

Marzano check - circle: ① ② ③

04 Plan a fair test

In the cart-and-hanger setup, you will change total moving mass while keeping the driving force the same.
1. **Label what you change** and what you measure.
2. **Name two controls**.
3. Why would adding mass to the hanger spoil this comparison?

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Record measurements honestly

If equipment is available, follow the teacher's setup and **record three acceleration trials** at each of two total masses. Otherwise use the supplied example data in **3** and **label it supplied example data**. **Compare the results**.

Support: My answer is ____. The two values that support it are ____.

Sentence starter: Both … . They differ in … because … .

Total M	Trial 1	Trial 2	Trial 3
____	____	____	____
____	____	____	____

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **4 kg** sign hangs still from one vertical cord. Use g=**10 N/kg**. **Name both forces**, their sources, magnitudes and directions.
Draw the forces on the chosen object and **label their sources**.

Sentence starter: In my work, I used … to show … . I labeled it … .

Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

At the same net force, a **1 kg** system accelerates at **6 m/s²**.
1. **Predict the acceleration of a 2 kg** system.
2. **State what must stay the** same for this prediction to work.

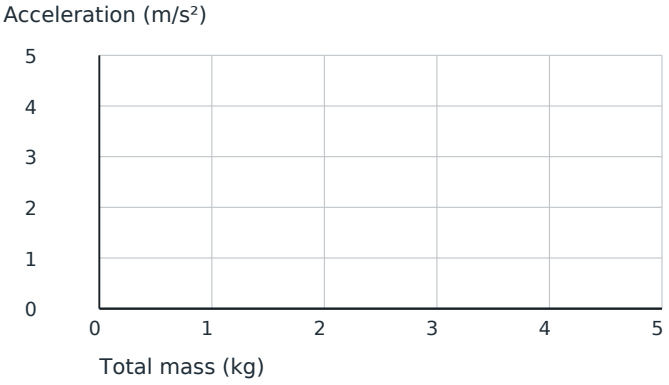
Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Return to the worked example in problem **2**. **Predict acceleration** for a **4 kg** cart at the same net force. Then **describe a fair trial** that could challenge your prediction.

Sentence starter: I predict … . I used … to make this prediction because … .



Marzano check - circle: ① ② ③